

# Shreyas Gupta

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## PROFESSIONAL SUMMARY

AI Research Engineer pursuing Integrated M.Tech in Artificial Intelligence (graduating May 2027). Hands-on experience building end-to-end AI systems, designing distributed architectures, deploying low-latency AI inference pipelines, and developing AI agents with Python and PyTorch. Proven track record in foundational model integration, AI system architecture design, and collaboration with research teams. Winner of Agentic AI Hackathon 2025 (NielsenIQ & TCS). Passionate about AGI research, distributed training systems, and scalable AI infrastructure.

## EDUCATION

**SRM Institute of Science and Technology** Chennai, India  
*Integrated M.Tech in Artificial Intelligence (CGPA: 8.29/10)* Aug 2022 – May 2027

- Coursework: Deep Learning, Neural Network Architectures, Distributed Computing, NLP, Computer Vision, Reinforcement Learning, AI Systems Design, High-Performance Computing

## EXPERIENCE

**AI/ML Engineer Intern** May 2026 – July 2026  
*Revenuxe* Bangalore, India

- Designed and built end-to-end AI system architecture including RAG pipelines, inference optimization, and multi-agent orchestration using Python, FastAPI, and LangChain across two AI products.
- Optimized AI inference systems for low-latency serving, implementing caching strategies and async request handling to reduce response time under concurrent load.
- Collaborated with AI research teams to ensure feasibility of implementation, translating research prototypes into production-grade distributed systems.
- Migrated AI inference backends to AWS (EC2, Lambda, S3) and Dockerized model deployments, building highly available and scalable serving infrastructure.

**Cybersecurity Research Intern** Dec 2025 – March 2026  
*Indian Institute of Technology, Guwahati* Guwahati, India

- Designed modular system architecture with CLI, parser, validator, loader, queries, and UI modules for AI-powered vulnerability analysis platform.
- Implemented custom LadyBugDB embedded graph database with optimized traversal algorithms, handling large-scale function call graphs and control flow edges efficiently.
- Built end-to-end AI pipeline from data ingestion to inference, demonstrating full-stack AI system construction and operation experience.

**Open Source Contributor (AI Engineer)** April 2023 – Present  
*NgKore Foundation* Remote

- Built autonomous Python AI agents with PyTorch-based reasoning for automated monitoring, anomaly detection, and configuration management in distributed 5G network systems.
- Designed distributed agent architectures with fault tolerance and self-correction capabilities for high-availability deployments; technical discussions conducted in English with global maintainers.

**Business Strategy Intern** May 2024 – July 2024  
*To-Let Globe* Remote

- Built data processing pipelines using Python and SQL to analyze 10,000+ listings; designed Power BI dashboards for real-time analytics.

## RESEARCH PROJECTS

**BharatGreen AI** | *Python, FastAPI, Next.js, TypeScript, NVIDIA NIM* Jan 2026 – March 2026

- Designed AI system architecture integrating foundational model inference (NVIDIA NIM) with distributed data pipelines for energy forecasting and carbon optimization.
- Implemented low-latency inference serving with model optimization, batch processing, and async request handling for real-time AI predictions.
- Developed scalable multimodal data pipeline handling temporal, geospatial, and text inputs with PyTorch-based model components.

**Light Pollution Sentinel** | *Python, Next.js, PostgreSQL, OpenAI, Docker* July 2025 – Sept 2025

- Built end-to-end AI system from satellite data ingestion to model inference and alerting across 742 districts, demonstrating full AI system construction and operation.
- Designed highly available system architecture with automated failover, health monitoring, and containerized deployment using Docker.
- Optimized inference pipeline for throughput and latency, processing large-scale NASA satellite datasets efficiently.

## AWARDS & ACHIEVEMENTS

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|---------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------|
| <b>Winner – Agentic AI Hackathon 2025 (Certificate)</b><br><i>NielsenIQ &amp; Tata Consultancy Services (TCS) – ALPS (Live)</i>             | Sep 2025<br><i>Chennai, India</i>           |
| <b>Member – Indian Society for Technical Education (ISTE) (Certificate)</b><br><i>SRM Institute of Science and Technology (SM No. 2258)</i> | Mar 2025 – Present<br><i>Chennai, India</i> |

## TECHNICAL SKILLS

**Core Languages:** Python, C/C++, Java, SQL (PostgreSQL), JavaScript, TypeScript  
**AI/ML Frameworks:** PyTorch, TensorFlow, LangChain, Hugging Face Transformers, OpenAI API, NVIDIA NIM, scikit-learn, NLTK  
**Distributed Systems:** Distributed training frameworks, model parallelism, data parallelism, async inference serving, low-latency optimization  
**Infrastructure & DevOps:** AWS (EC2, Lambda, S3), Docker, Linux, Git, GitHub, CI/CD, containerized deployments  
**System Architecture:** Low-latency inference systems, high-availability design, microservices, REST APIs, FastAPI, Flask, Next.js  
**Research Tools:** Jupyter Notebook, LaTeX, Weights & Biases, CUDA (basic), VS Code

## CERTIFICATIONS

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|------------------------------------------------------------------------------------------|---------|
| <b>TATA GenAI Powered Data Analytics Job Simulation</b> , Forage (Certificate)           | Oct '25 |
| <b>Deloitte Technology Job Simulation</b> , Forage (Certificate)                         | Jun '25 |
| <b>Natural Language Processing</b> , NPTEL (Certificate)                                 | Apr '25 |
| <b>Competitive Programming (Certificate of Excellence)</b> , Coding Ninjas (Certificate) | Jun '24 |
| <b>Programming in Java</b> , NPTEL (Certificate)                                         | Oct '23 |